A–B. Set Language, Constructions, and Regions

Elements, subsets, and equality

$x \in A$ means x is an element of A; $A \subseteq B$ means every element of A belongs to B. A proper subset $A \subset B$ also requires $A \neq B$. Sets are equal exactly when $A \subseteq B$ and $B \subseteq A$. Order and repetition do not matter inside a set.

Roster and set-builder notation

Roster notation lists elements; set-builder notation states a domain and a property. For example,

$$\{x \in \mathbb{Z} : -2 \leq x < 3\} = \{-2, -1, 0, 1, 2\}.$$

Read the colon as "such that." Always enforce the declared domain.

Power sets and Cartesian products

$\mathcal{P}(A)$ is the set of all subsets of A. If $|A| = n$, then $|\mathcal{P}(A)| = 2^n$. The empty set and A itself are included.

$$A \times B = \{(a, b) : a \in A \text{ and } b \in B\}.$$

Ordered pairs preserve order; generally $(a, b) \neq (b, a)$.

Similar example: a finite construction

Problem. Let $S = \{x \in \mathbb{Z} : |x| \leq 2\}$ and $T = \{-1, 1\}$. Write S in roster notation, decide whether $T \subseteq S$ and $T \in S$, and list every element of $\mathcal{P}(T)$.
Worked solution. Solve $|x| \leq 2$ in the integer domain to obtain $S = \{-2, -1, 0, 1, 2\}$. Both elements of T occur in S, so $T \subseteq S$. However, S contains integers, not the set T, so $T \notin S$. List subsets of T by size: zero elements, one element, then two elements. The count $4 = 2^2$ checks the result.
Answers.

$$S = \{-2, -1, 0, 1, 2\}, \quad T \subseteq S, \quad T \notin S,$$
$$\mathcal{P}(T) = \{\varnothing, \{-1\}, \{1\}, \{-1, 1\}\}.$$

Set operations

Relative to universe $\mathfrak{U}$,

$$A \cup B = \{x : x \in A \text{ or } x \in B\},$$
$$A \cap B = \{x : x \in A \text{ and } x \in B\},$$
$$A \setminus B = \{x : x \in A \text{ and } x \notin B\},$$
$$\overline{A} = \mathfrak{U} \setminus A.$$

Membership formulas and set formulas describe the same regions.

Translate logic into a Venn region

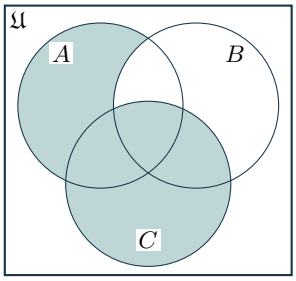
Replace $x \in A$ by A, $\vee$ by union, $\wedge$ by intersection, and $\neg$ by complement. First eliminate conditionals:

$$P \to Q \equiv \neg P \vee Q, \qquad P \leftrightarrow Q \equiv (P \wedge Q) \vee (\neg P \wedge \neg Q).$$

Then shade exactly the resulting region.

B.1: union and difference in a Venn region

Problem. Let $A, B, C \subseteq \mathfrak{U}$. Shade exactly those $x \in \mathfrak{U}$ for which $(x \in A \setminus B) \vee (x \in C)$ is true. Write the region using set operations.
Worked solution. The first disjunct requires membership in A but not B; the second requires membership in C. "Or" includes either or both, so shade all of C together with the part of A outside B.



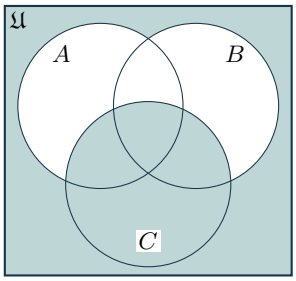
Answer. The shaded set is $(A \setminus B) \cup C$.

B.3: a conditional Venn region

Problem. Let $A, B, C \subseteq \mathfrak{U}$. Shade exactly those $x \in \mathfrak{U}$ for which $(x \in A \cup B) \to (x \in C)$ is true. Write the region using set operations.
Worked solution. Eliminate the implication:

$$\neg(x \in A \cup B) \vee (x \in C).$$

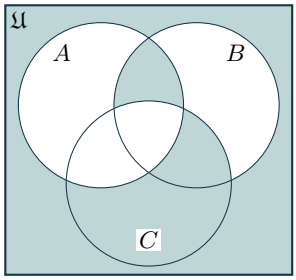
It is true outside $A \cup B$ and everywhere in C. It is false only inside $A \cup B$ but outside C.



Answer. The shaded set is $\overline{A \cup B} \cup C$.

B.5: a biconditional Venn region

Problem. Let $A, B, C \subseteq \mathfrak{U}$. Shade exactly those $x \in \mathfrak{U}$ for which $(x \in A) \leftrightarrow (x \in B \setminus C)$ is true. Write the region using set operations.
Worked solution. A biconditional is true when both sides are true *or* both are false. The first case gives $A \cap (B \setminus C)$; the second gives $\overline{A} \cap \overline{(B \setminus C)}$. Shade both.



Answer. The shaded set is

$$[A \cap (B \setminus C)] \cup [\overline{A} \cap \overline{(B \setminus C)}].$$

Common errors

$\{x\} \in \mathcal{P}(A)$ is equivalent to $x \in A$, but $\{x\} \in A$ is a different claim. Include $\varnothing$ in every power set. Do not treat a Cartesian product as unordered.

A–B readiness

Can you move between roster and set-builder notation; distinguish $\in$ from $\subseteq$; build $\mathcal{P}(A)$ and $A \times B$; and convert a compound membership formula into a set expression and Venn region?

C. Relations as Subsets of Cartesian Products

Relations and plots

A relation from X to Y is any subset $R \subseteq X \times Y$. A plotted point at horizontal coordinate x and vertical coordinate y represents $(x, y) \in R$. The **domain of the relation** is the set of first coordinates that occur; its **range** is the set of second coordinates.

When a relation is a function

A relation $R \subseteq X \times Y$ is a function $X \to Y$ exactly when

$$\forall x \in X \; \exists! y \in Y \; ((x, y) \in R).$$

Every input must have exactly one output. An input with no point or with two points proves the relation is not a function.

Injective and surjective

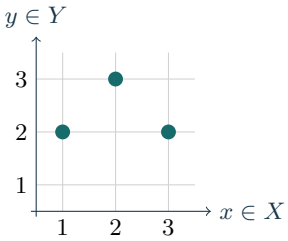
For a function $f : X \to Y$:

$$\begin{aligned} \text{injective}: &\quad f(x_1) = f(x_2) \Rightarrow x_1 = x_2, \\ \text{surjective}: &\quad \forall y \in Y \; \exists x \in X \; (f(x) = y), \\ \text{bijective}: &\quad \text{both injective and surjective.} \end{aligned}$$

On a finite plot, injective means no two inputs share an output; surjective means every codomain value is hit.

Similar example: reading an actual relation plot

Problem. Let $X = Y = \{1, 2, 3\}$. The plotted points are exactly the members of $R \subseteq X \times Y$. List R as ordered pairs. Decide whether it is a function $X \to Y$; if so, classify it as injective, surjective, both, or neither. Justify each answer.



Worked solution. Read each point as (horizontal coordinate, vertical coordinate): $R = \{(1, 2), (2, 3), (3, 2)\}$. Every input has exactly one output, so this is a function. Inputs 1 and 3 both map to 2, so it is not injective. Codomain value $1 \in Y$ is never hit, so it is not surjective.
Answer. $R = \{(1, 2), (2, 3), (3, 2)\}$ defines a function that is neither injective nor surjective.

Read quantified plot statements

$$\begin{aligned}
\forall x \exists y\, R(x,y) &: \text{ every column has a point,}\\
\forall x \exists! y\, R(x,y) &: \text{ every column has exactly one,}\\
\forall y \exists x\, R(x,y) &: \text{ every row has a point,}\\
\forall y \exists! x\, R(x,y) &: \text{ every row has exactly one,}\\
\exists x \forall y\, R(x,y) &: \text{ some column is full,}\\
\exists y \forall x\, R(x,y) &: \text{ some row is full.}
\end{aligned}$$

The first variable is the horizontal coordinate in (x, y).

Similar example: quantified relation statements

Problem. Let $X = Y = \{1,2,3\}$ and $R = \{(1,2),(2,3),(3,2)\}$. Use $R(x,y)$ to mean $(x,y) \in R$, with $x \in X$ and $y \in Y$. Decide whether each statement is true or false:

$$\forall x \exists! y\, R(x,y), \quad \forall y \exists x\, R(x,y), \quad \exists y \forall x\, R(x,y).$$

Worked solution.

1. Scan each input column: each has exactly one point.
2. Row $y = 1$ is empty, so this universal statement has a counterexample.
3. Row $y = 1$ is empty, row $y = 2$ misses $x = 2$, and row $y = 3$ misses $x = 1, 3$. Thus no row contains all three inputs.

Answers. True, false, false, respectively. For a false universal, name a counterexample; to disprove an existential here, check every possible row.

Similar example: diagnosing a nonfunction

Problem. Let $X = Y = \{1,2,3\}$ and $S = \{(1,1),(1,3),(2,2),(3,3)\}$. Does S define a function $X \to Y$? If it does, classify it as injective or surjective. If it does not, identify an input that violates the function requirement.
Worked solution. Input 1 has two distinct outputs, 1 and 3. A function must assign *exactly one* output to every input; having at least one is insufficient.
Answer. S is not a function, witnessed by input 1. The requested function classification therefore does not apply.

Common errors

Do not confuse "exactly one output for each input" with "exactly one input for each output." A function may be many-to-one. Surjectivity is always relative to the declared codomain, not merely the values visible in a formula.

C readiness

Can you list a plotted relation as ordered pairs, evaluate each quantifier pattern, identify a missing or duplicated output, and classify a valid function as injective, surjective, both, or neither?

D. Functions, Images, and Preimages

Domain, codomain, range

In $f : X \to Y$, X is the domain and Y the codomain. The range is $f(X) = \{f(x) : x \in X\}$ and may be smaller than Y. The same formula can have different injectivity or surjectivity behavior when its declared domain or codomain changes.

Images and preimages of sets

For $U \subseteq X$ and $P \subseteq Y$,

$$f(U) = \{f(x) : x \in U\}, \qquad f^{-1}(P) = \{x \in X : f(x) \in P\}.$$

$f^{-1}(P)$ is preimage notation and does not require f to have an inverse function.

D.1: finite images, preimages, and classification

Problem. Let $A = \{1,2,3,4\}$, $B = \{a,b,c\}$, and define $f : A \to B$ by $f(1) = a$, $f(2) = b$, $f(3) = b$, $f(4) = c$. Find $f(\{1,3,4\})$ and $f^{-1}(\{a,b\})$. Classify f as injective, surjective, both, or neither, and justify the answer.
Worked solution. Evaluate the inputs $1, 3, 4$ to obtain a, b, c. For the preimage, collect every input whose output is a or b: these are $1, 2, 3$. Every codomain element is hit, so f is surjective. However, $f(2) = f(3) = b$ although $2 \neq 3$, so it is not injective.
Answers.

$$f(\{1,3,4\}) = \{a,b,c\}, \qquad f^{-1}(\{a,b\}) = \{1,2,3\}.$$

The function is surjective but not injective.

Proving or disproving injectivity

To prove injective, start with arbitrary x_1, x_2 and assume $f(x_1) = f(x_2)$; derive $x_1 = x_2$. To disprove it, give distinct inputs with the same output.

Proving or disproving surjectivity

To prove surjective, let arbitrary y lie in the codomain and solve $f(x) = y$ for an allowed x in the domain. To disprove it, identify one codomain element with no preimage.

D.2: an integer formula

Problem. Let $g : \mathbb{Z} \to \mathbb{Z}$ be $g(n) = n + 5$. Find $g(\{-1, 0, 2\})$ and $g^{-1}(\{4, 7\})$. Classify g as injective, surjective, both, or neither. Justify both properties.

Worked solution. Substitute the three inputs: $g(-1) = 4$, $g(0) = 5$, $g(2) = 7$. For the preimage, solve $n + 5 = 4$ or $n + 5 = 7$, giving $n = -1$ or $n = 2$.

For injectivity, let $n_1, n_2 \in \mathbb{Z}$ and assume $g(n_1) = g(n_2)$. Then $n_1 + 5 = n_2 + 5$; subtracting 5 yields $n_1 = n_2$. For surjectivity, let arbitrary $y \in \mathbb{Z}$. Choose $n = y - 5 \in \mathbb{Z}$; then $g(n) = (y - 5) + 5 = y$.

Answers.

$$g(\{-1, 0, 2\}) = \{4, 5, 7\}, \qquad g^{-1}(\{4, 7\}) = \{-1, 2\}.$$

The function is bijective (both injective and surjective).

Similar example: domain and codomain matter

Problem. Compare $h : \mathbb{Z} \to \mathbb{Z}$, $h(n) = n^2$, with $q : \mathbb{N}_0 \to \{m^2 : m \in \mathbb{N}_0\}$, $q(n) = n^2$, where $\mathbb{N}_0 = \{0, 1, 2, \ldots\}$. Classify each as injective, surjective, both, or neither, using witnesses or proofs.

Worked solution. For h, distinct inputs -1 and 1 give the same output 1, so injectivity fails. The codomain contains -1, but no integer square equals -1, so surjectivity fails.

For q, assume $q(n_1) = q(n_2)$ with $n_1, n_2 \geq 0$. Then $(n_1 - n_2)(n_1 + n_2) = 0$. Either $n_1 = n_2$ or $n_1 + n_2 = 0$; in the latter case both are zero, so again $n_1 = n_2$. Thus q is injective. For any codomain value y, the codomain definition supplies an $m \in \mathbb{N}_0$ with $y = m^2$. Choosing input m gives $q(m) = y$, proving surjectivity.

Answers. h is neither; q is bijective. The formula is identical, but the declared domain removes sign collisions and the codomain removes unattainable values.

Common errors

An image removes duplicate outputs. A preimage collects inputs, not output values. Never classify a function without reading both domain and codomain. Do not claim surjectivity by saying "the formula has outputs"; construct a preimage for an arbitrary codomain element.

D readiness

Can you compute images and preimages; distinguish range from codomain; prove or disprove injectivity and surjectivity with the correct witnesses; and explain how changing the domain or codomain changes the classification?

E. Prove or Refute Set and Function Claims

Element-chasing proof template

To prove $S = T$, prove both inclusions. For $S \subseteq T$:

1. Let $z \in S$ be arbitrary.
2. Unpack membership in S using definitions.
3. Apply the relevant set/function definitions.
4. Conclude $z \in T$.

Do not begin by assuming the desired equality.

E.1: image of a union

Problem. Let $f : X \to Y$ and $U, V \subseteq X$. Decide whether $f(U \cup V) = f(U) \cup f(V)$ always holds. If true, prove it using an arbitrary element; if false, give a finite counterexample and compute both sides.

Worked solution. For the forward inclusion, let $y \in f(U \cup V)$. Then $y = f(x)$ for some $x \in U \cup V$. Thus $x \in U$ or $x \in V$, giving $y \in f(U)$ or $y \in f(V)$. Hence $y \in f(U) \cup f(V)$. For the reverse inclusion, let $y \in f(U) \cup f(V)$. If $y \in f(U)$, choose $x \in U$ with $f(x) = y$; if $y \in f(V)$, choose $x \in V$ with $f(x) = y$. Either case supplies $x \in U \cup V$, so $y \in f(U \cup V)$.

Conclusion. True: the two inclusions prove equality.

E.2: image of an intersection

Problem. Let $f : X \to Y$ and $U, V \subseteq X$. Decide whether $f(U \cap V) = f(U) \cap f(V)$ always holds. If true, prove it using an arbitrary element; if false, give a finite counterexample and compute both sides.

Worked solution. Take $X = \{1, 2\}$, $Y = \{a\}$, $f(1) = f(2) = a$, $U = \{1\}$, and $V = \{2\}$. These are valid sets and a function of the required types. Since $U \cap V = \varnothing$,

$$f(U \cap V) = f(\varnothing) = \varnothing.$$

But $f(U) = f(V) = \{a\}$, so

$$f(U) \cap f(V) = \{a\} \cap \{a\} = \{a\}.$$

Conclusion. False in general: the computed sides differ. Distinct inputs may have the same output, which is why intersecting images can add an output not obtained from the intersection of inputs.

E.3: preimage preserves intersection

Problem. Let $f : X \to Y$ and $P, Q \subseteq Y$. Decide whether $f^{-1}(P \cap Q) = f^{-1}(P) \cap f^{-1}(Q)$ always holds. If true, prove it using an arbitrary element; if false, give a finite counterexample.

Worked solution. For arbitrary $x \in X$, unpack the preimage and intersection definitions:

$$\begin{aligned} x \in f^{-1}(P \cap Q) &\iff f(x) \in P \cap Q \\ &\iff f(x) \in P \text{ and } f(x) \in Q \\ &\iff x \in f^{-1}(P) \text{ and } x \in f^{-1}(Q) \\ &\iff x \in f^{-1}(P) \cap f^{-1}(Q). \end{aligned}$$

Conclusion. True. Every step works in both directions, so the chain proves both inclusions and hence equality.

E.4: product distributes over union

Problem. For arbitrary sets A, B, C, decide whether $(A \cup B) \times C = (A \times C) \cup (B \times C)$ always holds. If true, prove it using an arbitrary ordered pair; if false, give a finite counterexample.

Worked solution. Let (x, y) be an arbitrary ordered pair. By the Cartesian product, union, and distributive definitions,

$$\begin{aligned} (x, y) \in (A \cup B) \times C &\iff (x \in A \text{ or } x \in B) \text{ and } y \in C \\ &\iff (x \in A \text{ and } y \in C) \\ &\qquad \text{or } (x \in B \text{ and } y \in C) \\ &\iff (x, y) \in (A \times C) \cup (B \times C). \end{aligned}$$

Conclusion. True. The reversible membership chain proves equality.

E.5: a false cancellation claim

Problem. For arbitrary sets A, B, C, does $A \cup B = A \cup C$ imply $B = C$? Prove the universal claim or give a finite counterexample that satisfies its hypothesis but violates its conclusion.

Worked solution. Take $A = \{1, 2\}$, $B = \{1\}$, and $C = \{2\}$. Then $A \cup B = \{1, 2\} = A \cup C$, so the hypothesis holds. However, $B \neq C$ because $1 \in B$ but $1 \notin C$.

Conclusion. False. Union with A can hide differences already contained in A. A valid counterexample must check both the true hypothesis and the false conclusion.

Common errors

Examples do not prove universal claims. For a false equality, compute both sides of a concrete finite example. In an image proof, different witnesses may map to the same output; do not silently identify them.

Whole-Core readiness

Can you construct finite sets, translate Venn regions, interpret relations, compute and classify functions, and choose between an arbitrary-element proof and a concrete counterexample? A live 20-minute form selects one complete problem from each section A–E.